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TOPICAL REVIEW

A Comprehensive Review on Next-Generation Modeling and Optimization for Semiconductor Devices

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ABSTRACT The integration of physics-based modelling and artificial intelligence (AI) is transforming semiconductor device simulation, facilitating unparalleled precision, efficiency, and predictive power. Conventional semiconductor modelling is based on first-principles physics, including drift-diffusion equations, Boltzmann transport models, and quantum mechanical methods. Nonetheless, these methods frequently encounter computational constraints when tackling intricate nanoscale processes. Novel AI-driven approaches, including as deep learning, physics-informed neural networks (PINNs), and alternative modelling, provide innovative ways to address these difficulties. The article explores recent progress in the integration of AI with semiconductor device physics, highlighting hybrid methodologies that preserve physical interpretability while utilising data-driven insights. In response to these improvements, machine learning-assisted compact modelling (MLCM) has garnered considerable attention as an alternative to conventional white-box modelling. These opaque methodologies seek to deliver versatile modelling for intricate mathematical and physical events through the training of neural networks using empirical and simulated data. This facilitates the creation of a precise closed-form correlation between output attributes and input parameters associated with the fabrication process and device functionality. Primary applications encompass swift process optimisation, concise model formulation, and inverse design for advanced electronics. It addresses the benefits and constraints of AI-based modelling, emphasising prospective approaches for combining physics-driven and data-driven paradigms. This interdisciplinary synthesis aims to expedite semiconductor research and development, promoting more efficient and scalable device design methodologies.

INDEX TERMS Artificial intelligence, Boltzmann transport, compact modeling, deep learning, machine learning, semiconductor.

I. INTRODUCTION

Modelling semiconductor devices is essential for electronics design, facilitating efficient, precise, and predictive simulations for many devices, including MOSFETs, BJTs, and novel nanoelectronics. Conventional methods rooted in semiconductor physics provide a comprehensive understanding of device functionality. Nonetheless, when device complexity and design scales increase, these models encounter constraints in computing efficiency and scalability. Recent breakthroughs, especially in machine learning, present intriguing solutions to these difficulties. The article presents a summary of the latest advancements in device modelling, contrasting conventional physics-based techniques with machine learning methodologies and assessing their prospective effects on future semiconductor technologies. Design Technology

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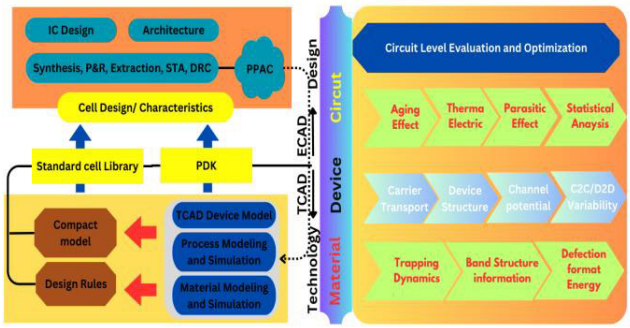

FIGURE 1. Schematic representation of the DTCO process [2].

Co-Optimization (DTCO) is a synergistic methodology in semiconductor design that simultaneously enhances technology and design to improve performance, power efficiency, and area (PPA) [1]. DTCO facilitates semiconductor manufacturers in advancing scaling limitations and efficiently fulfilling system requirements by aligning process technology breakthroughs with design strategies. In technology-based design co-optimization (DTCO) flows, semiconductor device modelling acts as a bridge between circuit design and device fabrication, as shown in Fig. 1 [2]. A number of crucial reliability concerns, such as quantum mechanisms, parasitic effects, instability (BTI), bias temperature, heat carrier degradation (HCD), kink impacts, and leakage of gate terminal, must be included in TCAD simulations in order to perform a precise assessment of the chip technology illustrated in TCAD. Thus, a remarkable TCAD model to SPICE model serves as the essential connection between the cutting-edge technology and integrated circuit design, incorporating many elements, including bias voltage, device shape, AC, DC, temperature, noise, and radiofrequency.

Physics-based TCAD models rely on well-established principles such as the drift-diffusion and Poisson equations, which describe carrier movement, charge distribution, and electrostatic interactions within devices. These models, which include the Shockley diode and MOSFET compact models, are extremely accurate and useful in understanding device behaviour. Numerous studies have been conducted to enhance the computational efficiency and predictability of TCAD simulations through the development of sophisticated physical models beyond drift diffusion (DD) frameworks and high-performance computing [4]. The drift-diffusion model and the Poisson equation are key equations that serve as the foundation for analytical and numerical solutions to device behaviour [5].

Drift-Diffusion Model: This model describes how carriers (electrons and holes) move in the presence of electric fields and concentration gradients. It is crucial for modelling current flow in devices such as MOSFETs.

Poisson's Equation: This equation is crucial for determining the electrostatic potential of devices, as it establishes the relationship between the and the electric field and charge distribution inside the semiconductor.

A. NUMERICAL SIMULATION MODELS

Numerical models solve complicated physical problems with great precision by using finite element and finite difference techniques [6]. These techniques are used by software such as Silvaco and Sentaurus TCAD, which allows for sophisticated modelling at the cost of increased processing power. These simulations can be too time-consuming for large-scale integration, but they are frequently necessary for the prototyping of new devices.

B. CONVENTIONAL PHYSICS-BASED COMPACT MODELS

Compact models are appropriate for circuit-level simulations because they endeavour to strike a balance between computational efficiency and accuracy. They use approximations and simplified formulae to depict semiconductor devices [7]. Although BSIM and other compact models are essential for designing digital and analogue circuits, their use for sophisticated devices with distinct geometries or quantum effects is occasionally limited by their reliance on simplifying assumptions [8]. The lightning-fast development of SPICE and MOS technology over the last half-century has allowed for a progression from simpler models of small metal-oxide-semiconductor field-effect transistors (MOSFETs) to the more sophisticated models utilised today [9], [10].

The swift progression of semiconductor technology has rendered the TCAD to SPICE methodology essential for connecting physical device simulations with circuit-level analysis. Recently, artificial intelligence (AI) has been included into this workflow to streamline the historically intricate and prolonged process of converting Technology Computer-Aided Design (TCAD) simulations into SPICE models, which are crucial for precise circuit modelling [11], [12]. The TCAD-to-SPICE process is essential in semiconductor design, enabling the development of SPICE models that depict the electrical characteristics of devices simulated in TCAD environments. Traditionally, this involves the manual extraction and fitting of parameters in order to ensure that SPICE models appropriately reflect TCAD-simulated device features. As devices downsize to nanoscale dimensions and structures grow more intricate, this technique has become progressively tedious and time-consuming. The recent integration of AI into this system resolves these challenges by automating parameter extraction, enhancing accuracy, and expediting model generation.

AI technology was recently developed to accelerate the development of the TCAD-to-SPICE technique. Data driven modelling, which includes look-up tables (LUT) and machine learning assisted compact modelling (MLCM) [13], is a black box technique that provides IC designers with detailed physical parameters, in contrast to physics-driven modelling. This approach avoids computing complexity but is devoid of physics based parameters, which impacts the convergence of simulations of VLSI. The table-based model, sometimes referred to as LUT, is typically utilised for the quick integration of observed device data into circuit models [14].

C. TRAINING METHODOLOGIES FOR MACHINE LEARNING-BASED COMPACT MODELS (MLCMS)

Although Machine Learning-based Compact Models (MLCMs) present an effective alternative to conventional equation-driven models, their effectiveness is predominantly dependant upon the quality and methodology employed in the generation of training data. MLCMs are generally trained utilising datasets obtained from systematic parameter sweeps, design-of-experiment (DoE) simulations, or Technology Computer-Aided Design (TCAD) outputs. In parameter sweep methodologies, device inputs including gatelength, oxide thickness, and bias conditions are systematically altered to provide a comprehensive array of output responses (e.g., I_D–V_G, I_D–V_D, capacitance curves). Design of Experiments (DoE) methodologies utilise statistical sampling to more effectively explore the design space, particularly advantageous when managing many variables or costly simulation executions. Datasets generated by TCAD are extensively utilised for training MLCMs due to their provision of high-resolution, physics-based data without the need for expensive fabrication, and their capability to simulate edge cases and rare phenomena, including short-channel effects, mobility degradation, and variability from random dopant fluctuations. Certain research utilise hybrid datasets that integrate TCAD simulations with empirical data to enhance real-world accuracy and model resilience. The training method generally involves correlating input-output pairs to substitute models, like neural networks, support vector machines, or Gaussian processes, which thereafter act as substitutes for physics-based models in SPICE or circuit simulators. Nonetheless, maintaining physical consistency and generalisation across technological nodes presents an ongoing problem, frequently necessitating meticulous feature engineering, regularisation, and the integration of physics-based restrictions during the training process.

The successful fusion of AI methods with semiconductor device physics has been shown in a number of recent research, leading to enhanced modelling precision, simulation effectiveness, and design optimisation:

A PINN-based compact modelling framework for nano-scale FETs was presented by Yang et al. [91] as an example of physics-informed neural networks for FETs. During learning, the trained neural network complied with Poisson's and drift-diffusion equations. The model allowed for the real-time prediction of I–V characteristics while providing a much lower computing cost and achieving near agreement with TCAD results. ML-Assisted Multi-Objective Optimisation in 2D FETs: Lin et al. [92] optimised important parameters in TMDC and black phosphorus FETs using Pareto-based machine learning. The project demonstrated how AI can speed up DTCO by successfully meeting the 2025 and 2028 IRDS targets by combining machine learning (ML) with a physics-based simulation backend.

Surrogate Modelling for Process-Aware FinFET Compact Models: Yalung et al. [93]replaced TCAD simulations in compact model extraction for FinFETs with a physics-aware neural network. By including established patterns and limitations into the network, the surrogate model preserved physical interpretability while capturing device behaviour across several process corners. AI-Based Reliability Modelling for Memory Devices: Razouk et al. [94] predicted endurance deterioration and failure modes in resistive memory arrays using long short-term memory (LSTM) networks. Accurate life-cycle prediction was made possible by the model's incorporation of domain-specific knowledge on filamentary switching and training using time-series current-voltage data.

A generalised technique for characterising process changes in LUT-based models was provided for circuit-level analysis, encompassing the description of I–V. The model is sensitive to data, and its accuracy is significantly affected by interpolation techniques, volume of data, and noise levels. The LUT, derived from an experimentally measured dataset and lacking physical explanation, demonstrates limited predictive capability, indicating its restricted applicability to pathfinding studies, particularly as the process development remains in its initial phases. The Artificial Neural Networks (ANN) based model is another data-driven compact modelling technique that replaces tables and interpolation techniques [15], [16]. It may be more scalable, generic, and smoother than a LUT model.

D. MACHINE LEARNING IN SEMICONDUCTOR DEVICE MODELING

With the limitations of physics-based approaches, especially for nanoscale devices and large datasets, machine learning (ML) methodologies have gained traction as a viable solution. ML models can tackle huge amounts of data and complex relationships that are challenging for traditional models [17], [18]. Data-driven machine learning models rely on training datasets derived from simulations or experimental measurements. Neural networks, Support vector machines in short (SVMs), and Gaussian processes have all been effectively used in semiconductor modelling to anticipate device behaviour with high accuracy and minimal reliance on physical approximations. Deep learning, particularly with designs such as RNNs (Recurrent neural networks) and CNNs (Convolutional neural networks) has shown the ability to predict with more accuracy across a wide range of device parameters [19], [20]. Deep learning applications include automatic feature extraction from raw data, anomaly detection in device behaviour, and transfer learning, which allows models to adapt to new device types with minimal retraining. Hybrid models combine the features of physics-based models and machine learning methodologies to achieve a balance of physical understanding and computational efficiency.The fig 2 summarises the benefits and constraints of each methodology, emphasising the complementary roles of conventional and ML-driven methodologies in meeting the different objectives of semiconductor device modelling For example, a hybrid strategy could utilise a classic compact model for base-level prediction and an ML model

to modify forecasts for specific conditions. This method improves model accuracy while keeping interpretability, reducing dependency on big training datasets.

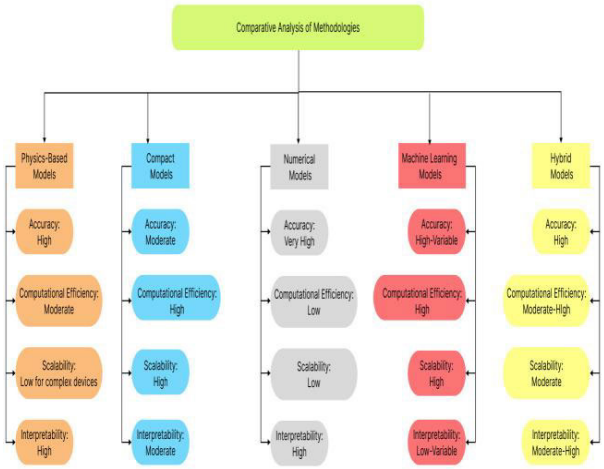

FIGURE 2. Schematic representation of the DTCO process [2].

II. DEVELOPMENT OF DEVICE PHYSICS ENGINE MODEL

A. OVERVIEW OF CONVENTIONAL TCAD MODELS

TCAD has undergone over seventy years of evolution since the introduction of the PN hypothesis analysis in 1949 (Fig. 3) [2]. Bell Labs, during the 1960s. IBM and Bell Labs conducted simulations in the early 1960s to simulate the behaviour of the diode in a specific device configuration. In the year 1964, Gummel introduced a self-consistent recursive technique for calculating single dimensional transistor simulations [21].

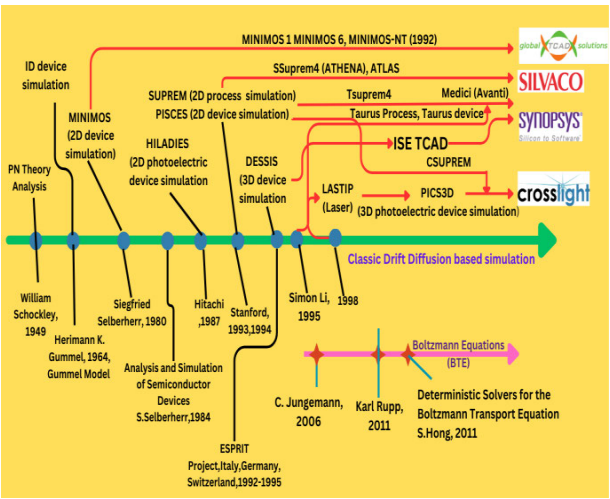

FIGURE 3. Evolution of TCAD [2].

The Gummel Poon model, introduced by Gummel and Poon in 1970, established the framework for the the triode model [22]. In 1975, Cottrell and Buturla introduced a 2- dimensional finite- element cross analysis approach [23] to enhance the algorithmic design for multi-dimensional device simulation tools, in light of the advancements in field-effect-controlled devices. Due to the limitations of one-dimensional simulations, which were unable to address complex coupling concerns arising from device shape scaling, Siegfried Selberherr in 1970s created the MINIMOS tool for the analysis of 2-dimensional FET's [24]. In the 1980s, three-dimensional device simulation tools such as MINIMOS 5 and MINIMOS 6 were developed. This advancement leads a vital role in comprehending and mitigating the short channel impact and optimising the doping profiles as a result of device miniaturisation. Likewise, D. A. Antoniadis developed the SUPREM family of tools for implementing integrated circuit modelling between in the mid-1980s [25]. while Pinto produced the PISCES IIB, a multi -dimensional device simulator [26]. Over the past two decades, the increasing complexity and high expenses associated with industrial production processes have made it crucial to prioritise the investigation of device structure design and performance optimisation with the assistance of TCAD.

Early development of TCAD led to the emergence of various popular TCAD programs, such as Global TCAD Solutions, Silvaco TCAD, Crosslight TCAD and Sentaurus TCAD. These tools have played a significant role in increasing the density through Design Technology Co-Optimization (DTCO), as shown in Fig 4 [27]. By 2023, by intensive scaling, it is possible to achieve a chip with hundreds of billions of transistors.

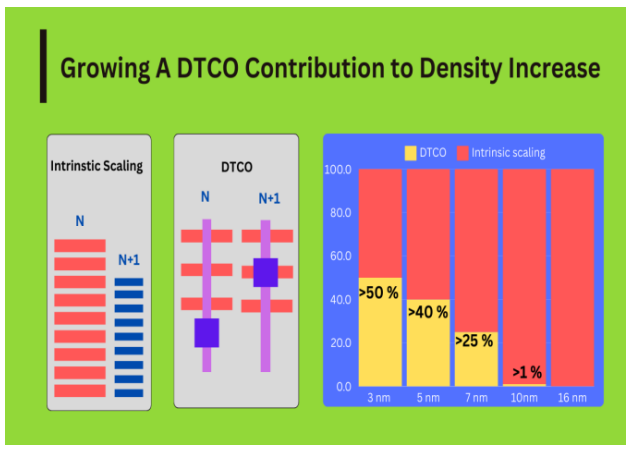

FIGURE 4. Increasing DTCO contributions drive advancements in gate density, surpassing traditional "intrinsic" scaling methods [27].

Also, there is a need to further investigate high-speed device modelling algorithms. Nevertheless, as device-scaling reaches the atomic level, standard TCAD faces the limitation of integrating quantum mechanisms, which might potentially result in a fundamental change from conventional device based physics to machine learning engines.

B. OVERVIEW OF CONVENTIONAL COMPACT MODELS

The Pao Sah model, first laid out in 1966, describes the drain to source current (I_{ds}) as: [28]

$$I_d = _\mu W Q_{id} . \frac{dV_c}{dy} \tag{1}$$

where V_c represents channel potential, Q_{id} represents charge density, μ denotes carrier mobility, and W represents channel width. The drift and diffusion behaviour are described by the Pao-Sah model, which is validated in all operational regions. It emphasises a crucial physical parameter referred to as the surface potential (ϕ_s), with the threshold voltage (V_{th}) which is defined as a particular value of the surface potential that delineates the operational mode. This model is exceedingly intricate, and the implicit relation necessitates an iteration algorithm to resolve ϕ_s that can only be resolved numerically. Consequently, since the late 1960s, numerous methodologies have been implemented to identify compact models for ECAD that are both computationally efficient and precise [29].

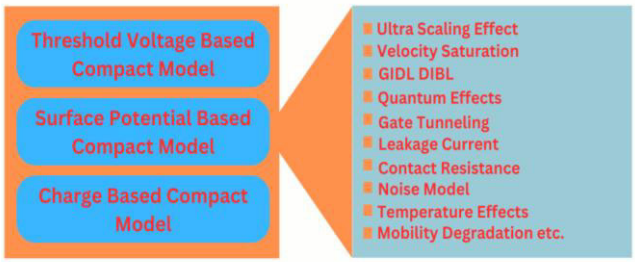

FIGURE 5. Schematic depiction of the fundamental model and external factors [2].

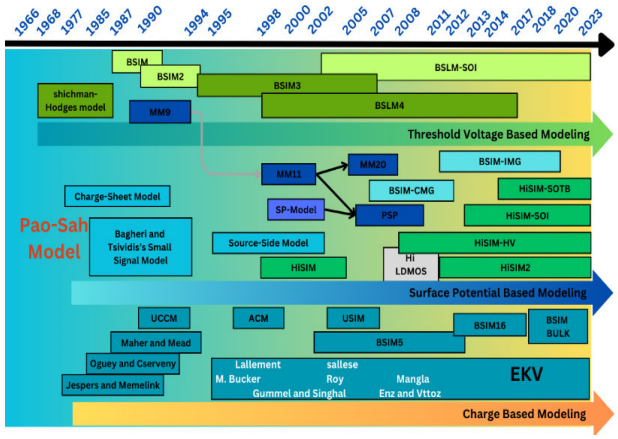

FIGURE 6. Advancement of popular compact based models: ϕs-based, Vth-based, & Q-based [2].

These models can be classified as surface potential-based (ϕ_s), threshold voltage based (V_{th}), and the charge based. (Figs. 5 and 6) [2].

1) MODELING BASED ON SURFACE POTENTIAL

The Pao-Sah equation is a current model that is based on the surface potential. Therefore, in order to maintain more physics data, the compact technique based on ϕ_s is favoured. However, there is a major challenge in determining an analytical surface potential at the channel terminal, and at the Drain-terminal ϕ_{sd} or Source-terminal ϕ_{ss}. The very first MOSFET charge sheet model, developed by Brews in 1977 [30], can be represented by a simple ϕs-based form [31].

$$I_d = \mu C_{ox}\frac{W}{L} \times \left[(V_{GS} - V_{FB} + 2V_T).\left(\varphi_{sd} - \varphi_{s_s}\right) - \frac{1}{2}(\varphi_{sd}^2 - \varphi_{s_s}^2)\right] \tag{2}$$

The benefit of such a model lies in its proximity to the underlying mechanics of the devices. In other words, the charges at terminal, currents, and derivatives depend on the ϕ_s. Researchers make significant advancements in solving implicit equations directly by including precise ϕs, resulting in substantial improvement during the 1980s., Bagheri and Tsividis in 1985 [32], [33] introduced a reliable technique using the Schroder series method. This method is widely utilised in current thin film transistors. In 1995, a source side model was published that effectively demonstrated the transition between several working modes for silicon-on-insulator (SOI) devices.

This approach provides a balance between precision and effectiveness by circumventing the need to solve the drain-side ϕ_s. The Hiroshima University STARC IGFET Model (HiSIM) [34] is distinct from the source-side-only model as it solves for two-side ϕs with both accuracy and efficiency. This model is designed to accommodate scaled devices and incorporates several significant physical effects. In order to address the extreme scaling behaviours in multi-gate devices, the surface potential technique has been utilised in BSIM models, including BSIM-IMG (Independent Multi Gate) [35] and BSIM CMG (Common Multi Gate) [36].

2) MODELING BASED ON THRESHOLD VOLTAGE

Analytically current can be determined by utilising V_{th} as the demarcation between different operational modes known as a V_{th} - based compact model. The benefit is that it removes the necessity for iterative calculations of ϕs and directly links the input voltage to the current equation, thus facilitating the prevalent use of the parabolic form. Mathematically it is formulated as:

$$I_d = K(V_{GS} - V_{th})^{2\cdot}(1 + l\,(V_{GS} - V_{GD}) \tag{3}$$

The MOSFET scaling rule was proposed in 1974 [37]. The ongoing trend of scaling leads to complex device physics and inaccuracies in the V_{th} based compact model. The Berkeley Short channel IGFET Model, often known as BSIM, was introduced in 1987 to enhance comprehension of short channel effects. It was subsequently followed by updated models, including BSIM2 and BSIM3 in the year 1990 and 1994 respectively. In addition, Philips Laboratories has introduced the MOS- Model 9 (MM9) [38], which is compatible with popular CAD programs. This model is characterised by well-behaved hyperbolic expressions and a reduced number of model parameters.

3) CHARGE-BASED MODELING

The inversion charge density (Q_{id}) at the two channel ends is used to generate the Q-based models. Like with ϕ_s, the task is to efficiently determine the charge density using an implicit equation. The rising complexity of V_{th} based models and the challenge of solving ϕs-based models analytically have led to the proposal of this alternate method.

A unified charge control model (UCCM) [39] that offers a framework for the relationship between charge densities and terminal voltages was proposed in 1990.

A current-voltage form based on UCCM is provided by:

$$I_d = q\mu(\alpha)\frac{W}{L}\left[\eta V_t(n_s - n_d) - \frac{\alpha}{2}\left(\eta_s^2 - \eta_d^2\right)\right] \quad (4)$$

where n_s and n_d denotes the electron concentration at the source and drain terminals of the channel region respectively.

Following that, other compact models based on charge have been described, including the unified charge control model (USIM) [40], the "Advanced Compact- MOSFET" model (ACM) [41], BSIM5, BSIM6, the EKV (Enz, Krummenacher, Vittoz) -MOSFET model [42], and others. However, for extensively scaled multi-gate systems, models based on ϕs are superior in precisely characterising the device's mechanics relative to alternative approaches.

III. COMPARISON OF TRADITIONAL PHYSICS BASED TCAD MODELS AND CONVENTIONAL PHYSICS BASED COMPACT MODELS

The semiconductor industry has long used physics-based models to simulate, predict, and optimise the behaviour of devices such as transistors and diodes. TCAD (Technology Computer-Aided Design) models and compact models are two popular approaches, with each serving a particular stage of the design and development process. This comparison focusses on their main features, strengths, and limitations. Table 1 represent a brief comparison between Traditional Physics based TCAD models and conventional Physics-based Compact models

IV. NAVIGATING THE CHAALENGES OF TRANSITIONING FROM TCAD TO SPICE IN THE POST-MOORE ERA

The physics based compact modelling is a method that focusses on solving Poisson's equation and is characterised by its white-box approach [43]. The core device characteristics of MOSFETs are mostly associated with the potential profile. It is crucial to establish an analytical relationship between MOSFET specifications and terminal voltages, assuming certain simplifications, in order to develop compact models. In addition, these models should adhere to fundamental criteria such as symmetry, scalability, continuity and efficiency while minimising the number of parameters used. However, in the post-Moore era, certain issues arise as obstacles to further advancement of physics-driven compact modelling methods, which align with the aforementioned needs.

A. QUANTUM RECTIFICATION AND ATOMIC PHYSICS

The principles of classical device physics must be broadened to encompass atomic physics as the size of Si-based transistors reaches its utmost limit. Quantum confined effects are observed in the highly scaled nanosheet gate all around FET (GAA-FET) [44], [45]. Therefore, the traditional Poisson's equation is linked with the Schrödinger equation. The thickness of a silicon-based ultra-thin channel determines the density of states function at its surface, as illustrated by SOIFETs. [46], [47], [48]. When analysing the geometric variance of multi-nanosheets, the fluctuation in threshold voltage will be a crucial but difficult factor to address in compact modelling. The necessity of incorporating quantum corrections into the standard model is crucial for those applications. Nevertheless, the intricate interdependent variables in Schrodinger equations pose a challenge in deriving analytical expressions by conventional compact modelling techniques. In order to address this problem, different approaches such as sub bands and Bohm's potential have developed to forecast the profiles of the carrier density and Bohm's potential caused by quantum processes.

B. OPTIMIZING NEAR-THRESHOLD OPERATION: INSIGHTS AND STRATEGIES

In today's near threshold circuit design, transition region models are becoming increasingly important. Nevertheless, the physical approach has a difficulty in accurately representing the change from the sub threshold to above threshold zone. A typical near threshold drain current model was proposed to enhance power dissipation optimisation and ensure performance, excluding the delay and power factors [49], [50] In the 2010s, researchers reported the use of near-threshold models [51], [52] using a fitting approach, inverse charge model, or important design specifications for accurately calculating energy and delay.

C. INCREASING COMPLEXITY

The complicated principles of physics result in the use of more abstract parameters for the purpose of fitting. Many fitting parameters are incorporated in empirical formulations, which results in increased complexity and poses challenges for parameter extraction. These intangible factors widen the space between the process and the design.

D. RELIABILITY CHARACTERIZATION

Reliability issues for advanced devices, such as predicting ultra-long -term aging, performance and statistical degradation, are becoming more challenging to address in device modelling. This is especially true for compact modelling, where cycle -to -cycle and device- to -device reliability are particularly important.

E. EXTREME ENVIRONMENT

MOSFETs are specifically designed to operate in adverse environments for specific applications, including quantum

TABLE 1. Comparison between traditional physics based TCAD models and conventional physics-based compact models.

Aspect	Traditional TCAD Model	Conventional Compact Model
Model Type	Numerical simulation based on the solution of fundamental physical equations (for example, Poisson's equation and the drift-diffusion equation).	Simple analytical or semi-empirical models that encapsulate device properties in a format that is suitable for circuit modelling.
Target Use	Complete device-level simulations for describing physical phenomena.	Optimised circuit-level simulations emphasising speed and scalability.
Complexity of Models	Extremely precise, encompassing detailed physical phenomena including quantum effects, hot-carrier effects, and strain effects.	Utilise simple equations using parameters derived from measurements or comprehensive simulations for approximation.
Scalability	Scalability is constrained; simulation time increases substantially with device dimensions and complexities.	Extensively scalable; efficient at simulating full circuits and complex systems.
Parameter Sensitivity	Extremely sensitive to material properties, shape, and doping profiles. Provides comprehensive details on how variations influence device performance.	Sensitive to electrical factors like mobility, threshold voltage etc. frequently calibrated to experimental data or derived from TCAD simulations.
Calibration	Requires specific physical characteristics, such as material specifications, mobility models, and doping profiles, some of which may require substantial experimental validation.	Parameter extraction from experimental measurements or TCAD simulations is essential for determining device performance. Calibration is easier but less accurate.
Industry Application	Used for device prototyping in R&D to test out new materials and device designs. Important for foundries which fabricate semiconductors.	Commonly used in industry-standard tools like SPICE to simulate circuits for IC design, from small analog circuits to enormous digital systems.
Example Tools	Sentaurus TCAD (Synopsys), Silvaco TCAD, ATLAS	BSIM Models (UC Berkeley), PSP Models, EKV Models
Device Architecture Support	Supports a variety of architectures, such as new and experimental devices such as quantum wells, FinFETs, and 2D materials.	Mostly focused on common designs (like MOSFETs), but with the right parameter sets, it can be changed to be compatible with new technologies.

computing, aerospace detecting and cryogenic high-performance computing, at temperatures as low as approximately 4.2 K. In regions with low temperatures, the consequences of structural disorders will be more apparent. Nevertheless, the majority of compact models in use today are not capable of extending to cryogenic conditions, necessitating revaluation of mobilities, threshold voltage.

Consequently, compact models in extreme environments still necessitate a significant amount of effort.

V. MACHINE LEARNING-AUGMENTED TCAD FOR SEMICONDUCTOR DEVICES

The ongoing advancement of semiconductor technology to sub-5 nm nodes and the incorporation of innovative materials and device topologies have presented substantial challenges to conventional modelling methodologies. Technology Computer-Aided Design (TCAD) has historically been fundamental to predictive semiconductor device simulation; nevertheless, its dependence on resolving intricate physical equations frequently results in prolonged simulation durations and constrained scalability when investigating extensive design spaces. The incorporation of machine learning (ML) techniques into TCAD workflows has emerged as a transformative solution to address these limits [53]. In a machine learning-enhanced TCAD framework, data-driven models are developed utilising comprehensive datasets derived from physics-based simulations, experimental outcomes, or historical design records. Upon training, these models can serve as effective surrogates for predicting electrical characteristics, including current-voltage (I–V) behaviour, threshold voltage, capacitance, and carrier distribution. By markedly diminishing the necessity for iterative numerical solutions of partial differential equations, machine learning models facilitate expedited investigation of process and design modifications.

A diverse array of machine learning algorithms, such as artificial neural networks (ANNs), decision trees, and Gaussian process regression (GPR), has been employed to approximate and expedite critical components of device simulation. Recently, deep learning methodologies and physics-informed neural networks (PINNs) have enabled the acquisition of intricate device behaviour while directly integrating fundamental physical rules into the training process. Physics-Informed Neural Networks PINNs enhance semiconductor modelling by including fundamental physical laws,

such as Poisson's equation and continuity equations, directly into the loss function during the training process. This ensures that the ML model not only responds to the data but also adheres to the fundamental principles of device physics, which is particularly advantageous in data-sparse or extrapolated contexts. PINNs are very effective for addressing PDEs that regulate charge transport, rendering them suitable for nanoscale and novel device architectures.

Physics-Informed Neural Networks (PINNs) integrate physical rules into the training procedure, enabling superior generalisation with reduced data relative to conventional neural networks. Although conventional networks often exhibit quicker training in data-abundant settings, Physics-Informed Neural Networks (PINNs) demonstrate superior performance in extrapolative tasks owing to their physics-based limitations. Nonetheless, PINNs may exhibit slower convergence and necessitate precise modifications, particularly for complex physical systems. Their power relies in simulating scenarios with less data while possessing an in-depth knowledge of the underlying physics.

This guarantees both precision and comprehensibility of the predictions, even when extending beyond the training domain.

A. ACCELERATING DRIFT-DIFFUSION-BASED DEVICE MODELING USING MACHINE LEARNING

Semi-classical simulation methodologies, especially those utilising drift-diffusion (DD) and hydrodynamic models, continue to be prevalent owing to its equilibrium between computational efficiency and physical fidelity. Nonetheless, as devices diminish to nanometre dimensions and include intricate materials or geometries, these models encounter computational limitations during comprehensive design space exploration. To resolve this, machine learning (ML) techniques have been included into semi-classical TCAD frameworks to create rapid surrogate models capable of predicting essential device parameters, including threshold voltage, drive current, and subthreshold slope. Artificial neural networks (ANNs) have been effectively trained on TCAD-generated datasets to accurately replicate current-voltage (I–V) and capacitance-voltage (C–V) properties, thereby greatly expediting parametric analysis and design optimisation procedures. Additionally, hybrid modelling approaches that integrate machine learning with differential equation solvers have been suggested to optimise potential distributions and carrier profiles while minimising the frequency of iterative partial differential equation solutions. The collaboration between machine learning and semi-classical models not only accelerates simulation speed but also facilitates real-time sensitivity analysis and inverse design activities, consequently enhancing the overall efficiency of contemporary device development workflows [55].

Deep Learning (DL): Deep learning models, particularly deep neural networks (DNNs), proficiently capture intricate, nonlinear correlations between device attributes (e.g., geometry, doping) and electrical responses (e.g., I–V characteristics). Deep learning facilitates swift alternative modelling, significantly diminishing simulation duration relative to conventional TCAD solvers. It facilitates high-dimensional design space exploration and assists in device optimisation and inverse modelling activities.

This has resulted in a significant reduction in the computational costs associated with predicting various electrical characteristics, such as electrostatic potential, IV and CV characteristics [48], [49], [50], threshold voltage, metal work function of the device and other performance metrices. In the same way, machine learning is effective in both the prediction of device characteristics and the optimisation of devices when it is combined with a multifaceted optimisation algorithm that meticulously evaluates the trade-off between electrical properties. A framework utilising Pareto active learning was proposed to optimise black phosphorene FETs and 2D transition metal dichalcogenide (TMDC) [56]. The framework, which incorporates multi-objective optimisation (MOO), successfully optimised the 2D FET to meet the technology nodes specified in the International Roadmap of Devices and Systems for the year 2025 & 2028. This optimisation was illustrated in Fig 7 [56].

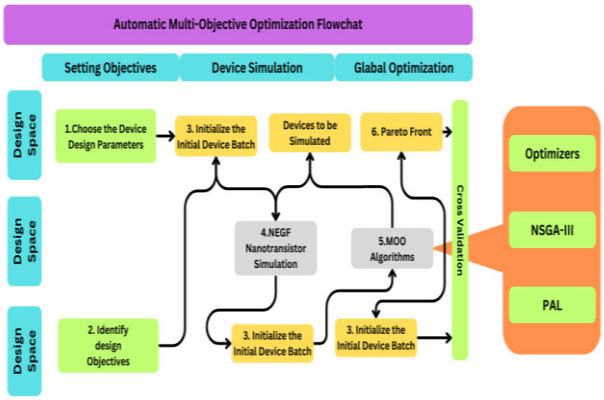

FIGURE 7. Proposed MOO framework integrated with ML [56].

B. ATOMISTIC COMPUTATIONS UTILISING MACHINE LEARNING

DFT-based ab-initio TCAD and Density functional theory (DFT) are effective methods for determining and examining the characteristics of materials in nano devices [57]. These methods are particularly useful when the size of the devices becomes extremely small and quantum effects become significant.

Nevertheless, the computing cost of DFT is considerable, as there is a compromise between precision and efficiency. Hence, new machine learning techniques are developed, wherein neural networks (NN) provide promising outcomes in capturing nonlinear properties and predictive significance. As an example, the Schrodinger equation can be solved with great efficiency and virtually precise results for molecules containing up to 30 electrons, outperforming the precision of the state-of-the-art version [2].

One more instance involves the use of neural networks as a wave-function ansatz in ab-initio calculations to forecast the dissociation process of the nitrogen molecule and hydrogen chain. The neural network configuration may also contribute to the computation. A recent work has shown that neurons can function as the tight-binding (TB) matrix elements in the Hamiltonian parameterisation of the TB model for energy band calculation [58], [59]. Various studies have reported that machine learning-enhanced density functional theory (DFT) is effective and accurate for atomistic modelling of devices. This includes forecasting atomic force interaction in phase change memory [41], calculating potential energy surfaces in Si-Ge alloy, and simulating surface reconstruction of the Si(111)-(7 × 7) surface. Quantum transport properties and device merits play a crucial role in simulations for actual TCAD applications aimed at developing the 5 nm node and beyond. Recent studies have employed the non-equilibrium Green's function (NEGF) method alongside machine learning to model quantum transport. This simulation includes the calculation of electrostatic characteristics. M. Burkle et al have introduced a simulation framework (Fig. 8) that combines neural networks (NN) and non-equilibrium Green's function (NEGF) approaches [60]. This framework can qualitatively predict the conductance of a larger system with high accuracy. Moreover, it achieves this with computational costs that are only a fraction of those associated with traditional first-principle methods. In addition, Y. Zhou et al. have recently introduced AD NEGF, which is the initial fully differentiable NEGF model for simulating quantum flow.

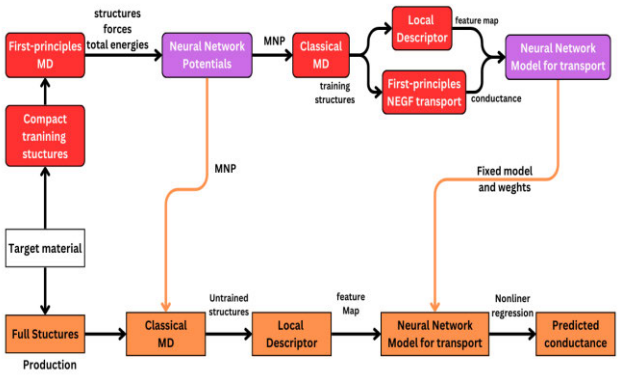

FIGURE 8. Proposed NN–NEGF computation framework by. Burke et al [60].

This model leverages the Pytorch deep learning framework [61] for numerical computations and efficiently computes the backward gradient using the proposed implicit layer approach. The examples discussed above show that the ML approach is a highly effective solution for overcoming the computational efficiency limitations of first-principles methods in device analysis and optimization.

VI. DEVELOPMENT OF A LIGHT WEIGHT MACHINE LEARNING MODEL FOR TCAD INTEGRATION

The advancement of compact and efficient machine learning models is essential in semiconductor device modelling, since it addresses computational speed with physical accuracy. These lightweight models are intended to enhance TCAD simulations by providing surrogate modelling capabilities, facilitating swift assessment of device performance indicators across extensive parametric ranges. Artificial neural networks (ANNs) have become the predominant machine learning architecture because of their ability to approximate nonlinear functions. An effectively designed ANN model can incorporate a continuous activation function that seamlessly represents behaviour throughout all operational zones of a device. Nonetheless, constructing such models poses considerable difficulties in choosing suitable input-output mappings and optimization algorithms for parameter extraction, particularly within the black-box modelling framework where interpretability is frequently constrained.

A. ANN-BASED MODELING FOR RF FETS, MICROWAVE DEVICES, AND HEMTS

Artificial Neural Network-Based Modelling for RF Field Effect Transistors, Microwave Devices, and High Electron Mobility Transistors. In high-frequency electronics, devices like RF FETs, microwave transistors, and High Electron Mobility Transistors (HEMTs) [10], [62], [63], [64] demonstrate intricate dynamic behaviour, encompassing parasitic effects, thermal feedback, and noise. Conventional empirical models frequently fail to adequately represent this complexity across temperature and frequency domains. To tackle this issue, initial applications of artificial neural networks—particularly multilayer perceptrons (MLPs)—were utilised to simulate DC characteristics, S-parameters, and noise figures of microwave transistors. Adjoint neural network approaches were introduced for sensitivity analysis, enhancing the trainability and generalisation of ANN models by utilising derivative information. The commercial sector subsequently developed measurement based neuro models like DynaFET and Neuro FET [65], [66]. Additionally, the MLCM technology was introduced to speed up parasitic aware RF circuit analysis by generating compact distributed circuit models quickly and accurately. Researchers have utilised the ANN approach to analyse various aspects of high electron mobility transistors (HEMT). These include studying current specification, noise characteristics and S-parameters in relation to temperature properties and frequency. An electro thermal model based on artificial neural networks was developed to address kink effects, self-heating and dynamic trapping [67], [68].

B. MACHINE LEARNING IN SCALED SILICON MOSFETS AND THIN-FILM TRANSISTOR (TFT) TECHNOLOGIES

As CMOS technology advances to sub-3 nm nodes, device physics becomes more complex due to quantum confinement, short-channel effects, and unpredictability. Modelling these behaviours in traditional TCAD frameworks frequently necessitates intricate meshing and sophisticated solvers, resulting in high computing demands. MLCM techniques

[7], [13], [15], [69], [70], [71], [72] present a viable alternative by utilising simulated or experimental datasets to create precise compact models capable of predicting DC, AC, and transientbehaviour.

Thin-film transistors (TFTs), especially those utilising oxide semiconductors and 2D materials, demonstrate operational principles that markedly differ from those of bulk MOSFETs, particularly in the context of future display and sensing applications. Their distinct surface potential distributions and movement traits necessitate specialised models. Machine learning has been utilised to develop generalised modelling frameworks capable of accommodating differences in material properties, channel geometry, and biassing schemes. Recent research has employed datasets produced by Non-Equilibrium Green's Function (NEGF) [73], [74], [75] solvers to develop surrogate models for these innovative gadgets. These data-driven models have been essential in investigating the performance boundaries of novel channel materials and device architectures, including IGZO, MoS2, and WSe2-based TFTs.

C. MLCM FRAMEWORK

A comprehensive methodology to develop MLCM is outlined, consisting of five essential processes: (1) determining parameters, (2) preparing the dataset, (3) selecting the appropriate algorithm, (4) coding in Verilog-A, and (5) simulating the circuit. These procedures are depicted in Fig 9 [2].

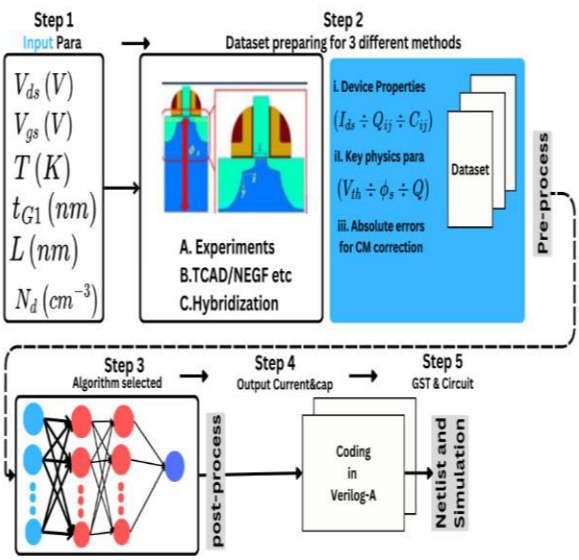

FIGURE 9. Process of developing an ML compact model [2].

The key factors consist of terminal voltages, temperature, length of the gate, gate-oxide thickness and doping profiles N_d. Typically, devices are evaluated under defined parameters, such as a particular temperature and precise gate lengths. There is a possibility that the MLCM model will over fit if the parameters are directly used as inputs. In order to prevent this problem, one can utilise Taylor expansion assisted modelling to represent the specific parameter dependence and thus lower the dimension of the ANN model. Datasets are fundamentally crucial in data based technology and could be acquired by experimentation, simulation, or a combination of both methods. The advancement of MLCM using experimental datasets is significantly constrained for advanced devices, let's say GAA-FETs, Fe-SOI-FETs and ultra-scaled vertical IGZO FETs. [76], [77], [78], [79], [80].

The IGZO compact model's description can be further enhanced by incorporating essential developments from recent surface-potential-based modelling endeavours. A significant compact model for a-IGZO TFTs is presented in (Zong et al., IEDM 2014) [89], which employs a surface potential-based approach that is based on the multiple trapping and release (MTR) theory. This model effectively captures the physics of disordered semiconductors, accounting for subgap trap states that influence charge transport and subthreshold behaviour. This is particularly important for applications such as RFID, where accurate modelling of low-voltage operation is essential.

(Guo et al., IEDM 2021) [90] expands the modelling framework to ultrashort channel lengths (sub-12 nm) by incorporating short-channel effects, quantum confinement, and gate dielectric scaling, building upon this foundation. The proposed model is validated against experimental data for a diverse range of oxide thicknesses and channel lengths, thereby demonstrating its scalability and strong correlation. This work emphasises the significance of physics-based modelling for aggressively scaled oxide semiconductor devices, particularly in the area of emergent DRAM and other low-power memory technologies.

The physical accuracy, scalability, and application relevance of the current IGZO compact model to both large-area electronics and aggressively scaled nano-electronic devices would be improved by incorporating these advancements.

Device simulators, such as TCAD or NEGF, are used to increase the amount of the dataset for these applications by varying different parameters. However, a common gap exists between experimental and simulated results, and transferring models trained on simulations to those trained on experiments remains a key challenge. Algorithms like transfer learning present potential solutions to this problem. Various MLCM approaches necessitate distinct datasets for training, which can be categorised into three types and illustrated in Figs. 10, 11,12, and 13.

i-type: The data is consistent with the outputs of the device. Experimental or simulated electrical characteristics are directly utilised or converted regardless of traditional CMs [81].

ii-type: The data pertains to the primary physics-based parameters of the devices that are employed in hybrid ML models. A physics-inconsistent issue must be considered in order to adhere to the fundamental principles. The primary challenge of CM, as it pertains to the three main classical models, is the acquisition of critical channel parameters (V_{th}/ ϕ_s / Q) in the scaling-down of advanced technology nodes by solving the Poisson's equation, even when coupled with the

Schrödinger equation. Consequently, the ii-type data-driven models are built upon the traditional method by substituting the intractable component.

iii-type: The data corresponds to the corrected term used for correcting the classical model.

iii-type data-driven models aim to rectify errors and enhance accuracy in industry standard models. They also offer an alternate approach for quantum-corrected modelling.

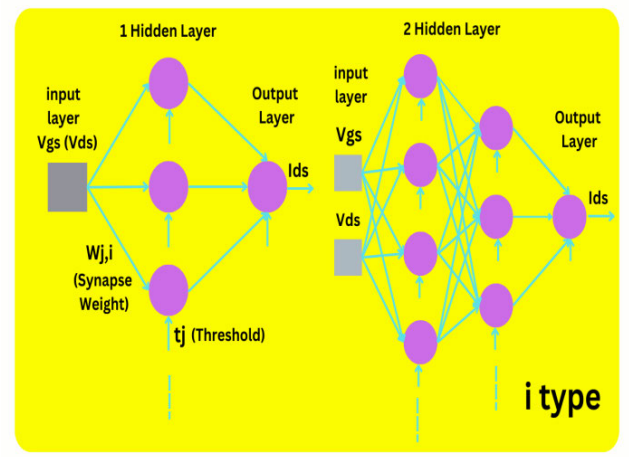

FIGURE 10. Voltage inputs and current outputs in feed-forward networks [81].

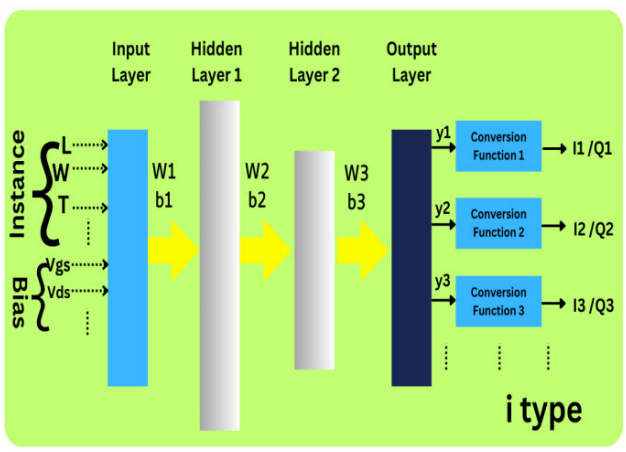

FIGURE 11. Conversion of output variables using ANN in [7].

1. DATA PRE-PROCESSING is crucial for enhancing the efficiency and accuracy of concise models. Typically, I_{kanar} covers a wide range of values, spanning many orders of magnitude. For example, even the ratio of I_{on} to I_{off} is approximately 1015 for IGZO transistors. During training processes, numerical difficulties, such as the weight attribution may arise. Proposed solutions such as initial MOSFET sizing, and logarithm I_{ds} / V_{ds} pre-processing, Latin hypercube sampling algorithms [83], [84], are suggested to address these challenges and enhance efficiency.
2. THE SELECTION OF AN ALGORITHM is essential for the development of MLCM. The backpropagation method of multilayer perceptron (MLP) has been extensively employed in the compact modelling of semiconductor devices [13]. Hyperbolic tangent, ReLu, sigmoid, softplus, softsign, ISRLU, ISRU, and Swish are among the activation functions that are currently available. In order to enhance the accuracy of training, hyperbolic tangent and sigmoid hybrid activation functions are implemented [6], [85].
3. VERILOG-A is regarded as the industry standard language for the implementation of compact based models in circuit simulators [84]. The ANN algorithm can be realised by utilising an automation script (such as Python [85]) to perform the transformation process, as illustrated in Table 1, as the Verilog-A language is not ideal for calculations of matrix. Consequently, matrix calculations are transformed into fundamental multiplications that are in line with Verilog-A.
4. SIMULATIONS are run using Verilog-A code. Nevertheless, simulations of circuits can encounter convergence issues with certain MLCM techniques, caused by high-order derivatives of terminal current. Thus, similar to conventional compact modelling approaches, it is necessary to conduct Gummel symmetry tests (GST) to evaluate the consistency and convergence. Until now, only a small number of papers (74, 103) have documented this test.

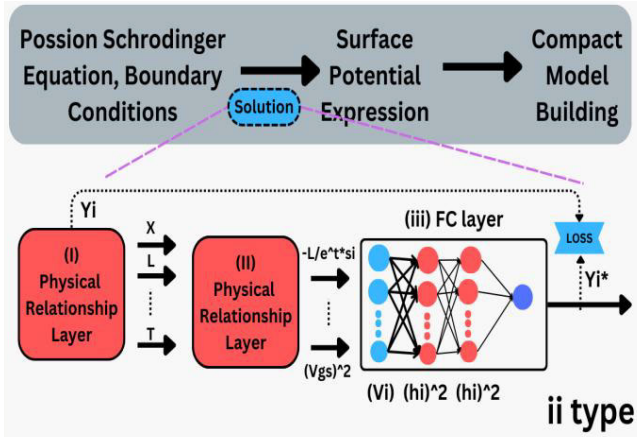

FIGURE 12. TCAD simulation in PRNN [7].

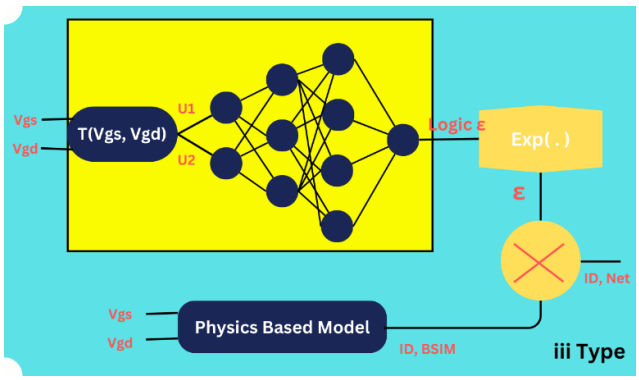

FIGURE 13. Neural network design using the core BSIM-CMG as the foundational physics-based model [82].

D. PHYSICS INTEGRATED TECHNOLOGY

As previously stated, a major issue with the data-driven modelling methodology is the presence of physical discrepancies. In accordance with fundamental physical principles, such as Ohm's law, when the voltage between the drain and source terminals, V_{ds}, is equal to zero, there should be no current flowing between them. Furthermore, the anticipated results should consistently converge beyond the voltage range utilised for training. In order to fulfil the aforementioned criteria, neural network models might be imbued with an understanding of physics by means of algorithmic optimisation as shown in fig 14 [2]. or incorporating past information.

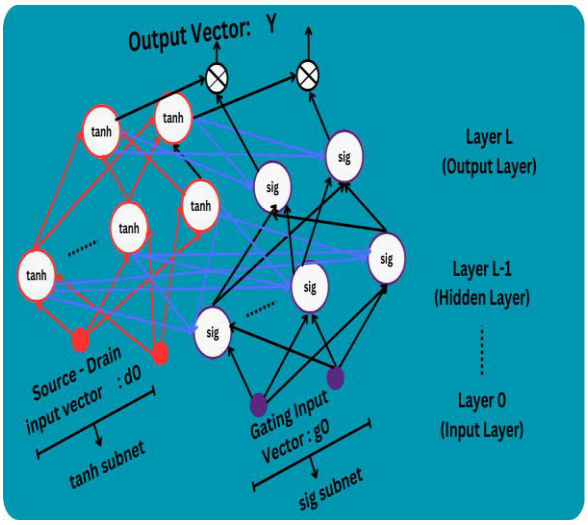

FIGURE 14. Technology with physics aware [2].

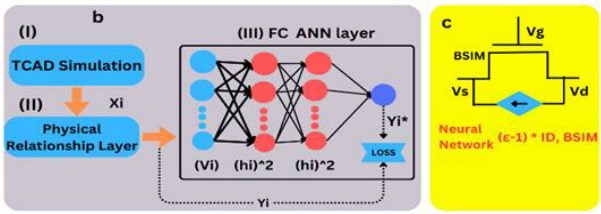

FIGURE 15. (a) Adapted from ref [85] (b) Adapted from ref [86].

1. ***Optimisingalgorithms*** improves the ability of neural network models to understand and incorporate physical principles. An effective method to accomplish this is by considering CM as a regression problem, resulting in the utilisation of the PINN algorithm. The operating principles of the device vary depending on the applied voltages, as observed from a physics-based perspective. Consequently, the NN design can be divided into two subnets: a tan h subnet with a source-drain input vector and a sig subnet with a gating input vector, as seen in Figs 14,15 (a) and (b). By enhancing the algorithm, the output features align more consistently with the fundamental principles of physics.For i-type NN models, a simple conversion considering an Ohm's law would be as follows:

$$I_d = I_0 V_{ds} 1_0^y \tag{5}$$

where y denotes the output of the ANN models. The current is compelled to zero at $V_{ds} = 0$ in this expression. Various physics-aware methods have been proposed in the literatures [87] and [88]

A physics-based neural network has been proposed for ii-type NN models, which are adapted from the traditional CM development process. This network includes a physical-relation layer to integrate fundamental physical prior knowledge and more effectively capture data nonlinearity. The physical layer is defined as follows:

$$\vartheta_i = [V_{GS}, V_{DS}, N_D, T, x, e^{\lambda \frac{x-L}{t_{Si}}}, VDS^{e\lambda \frac{-L}{t_{Si}}}, V_{Bi}(1 - e^{\lambda \frac{x-L}{t_{Si}}}), V_{GS}^2, V_{GS} - V_{DS}, \log\left(\frac{L_{DI} V_{GS}}{t_{Si}}\right) \tag{6}$$

A hybrid model can be represented as follows by employing trained surface potential:

$$I_{DS} = \mu_{effe} C_{ox} \frac{W}{L}\left[V_G(\varphi_{sd} - \varphi_{ss}) - \frac{1}{2}(\phi_{sd}^2 - \phi_{ss}^2)\right] \tag{7}$$

The surface potential near the source terminal is determined by the ANN:

$$\phi_{ss} = \varphi_s[V_{GS}, V_{DS}, T, x] - \varphi_s[0, 0, T, x] \tag{8}$$

$$\phi_{sd} = \varphi_{ss} + V_{DSx} \tag{9}$$

$$V_{DS}X = gV_{DS}^{\left(1+\frac{g_{V_{DS}}^m}{V_{Dsat}}\right)\frac{-1}{m}} \tag{10}$$

where $\phi_{sd} = \phi_s s$ and the current is zero at $V_{DSx} = 0$

Furthermore, this technique is more adaptable to various characteristic expressions, such as terminal charge/ capacitance, as indicated by:

$$Q_G = W \times Cox \int_0^l (V_G - V_{FB} - \varphi_s)([V_{GS}, V_{DS}, T, x])\, dx \tag{11}$$

$$Q_D = W \times Cox \int_0^l q_i \frac{y}{l} dx \tag{12}$$

$$Qs = W \times \int_0^l q_i \left(1 - \frac{y}{l}\right) dx \tag{13}$$

$$C_{MN} = -\frac{\partial Q_M}{\partial V_N}, C_{MM} = \frac{\partial Q_M}{\partial VM} \tag{14}$$

The correction function must fulfil the following equation for iii-type NN models, accounting for the zero-problem and symmetric devices:

$$I_{DS,BSIM}(V_{GS}, V_{GD}) \times \varepsilon(V_{GS}, V_{GD}) = -I_{DS,BSIM}(V_{GD}, V_{GS}) \times \varepsilon(V_{GD}, V_{GS}) \tag{15}$$

E. FEW-SHOT LEARNING AND TRANSFER LEARNING IN ADVANCED NODES

Process-aware simulations are crucial for ultra-scaled devices that are affected by manufacturing unpredictability, same to how traditional statistical compact models are vital. MLCM introduces statistical effects due to the enormous processing of data in advanced technology nodes. The distribution of high or low resistance states is crucial for analysing the performance of memory devices, especially in circuit analysis. These methods require obtaining the standard deviations of process parameters, and then the mean and variance are propagated through the artificial neural network (ANN). In the context of few-shot learning for advanced fabrication devices such as GAA-FET, transfer learning has been suggested as a solution for failure analysis in transistors beyond 3 nm in CMOS technology.

F. PARAMETER EXTRACTION ASSISTED BY MACHINE LEARNING

Parameter extraction is a vital and labour-intensive phase in the creation and implementation of compact models, particularly for sophisticated device designs like FinFETs, Gate-All-Around (GAA) FETs, and other three-dimensional geometries. Modern compact models, such BSIM-CMG, BSIM-IMG, and HiSIM2, possess numerous interdependent parameters, leading classic gradient-based or heuristic optimisation techniques to encounter challenges such as non-convergence, heightened sensitivity, and overfitting. The aforementioned restrictions have prompted the incorporation of machine learning (ML) methodologies into the parameter extraction process. A method utilising machine learning to extract parameters was presented to address this issue for the BSIM-CMG model, which is a highly effective approach for accurate calibration [89].

1) SURROGATE-BASED INVERSE MODELING FOR PARAMETER EXTRACTION

Unlike direct fitting methods, ML-driven extraction regards the compact model as a black box and learns the inverse mapping from specified electrical properties (e.g., threshold voltage, drain current, output conductance) to the model parameters. This is generally accomplished by supervised regression models, such artificial neural networks (ANNs), support vector regression (SVR), or Gaussian process regressors (GPR), which are trained on synthetic datasets produced by TCAD or SPICE-level simulations.

Modelling frameworks for semiconductor devices are frequently classified as **white-box, black-box, or hybrid based** on how they integrate data and physical knowledge. In order to capture the fundamental behaviour of electronic devices, white-box models are based solely on physics-based formulations and rely on analytical equations like Poisson's and drift-diffusion models. TCAD simulations and small models like BSIM and HiSIM are two examples. Although these models have strong extrapolation capabilities and high interpretability, they frequently necessitate a great deal of parameter extraction and may not be scalable or complex enough for contemporary nanodevices. Black-box models, on the other hand, lack any embedded physical insight and are solely data-driven, learning the mapping between input parameters and output characteristics. This group includes machine learning methods like support vector machines and artificial neural networks. Black-box models lack transparency and can produce nonphysical results when used outside of the trained domain, despite their efficiency and suitability for approximating complex, nonlinear behaviour. By integrating physical rules into data-driven structures, hybrid models bridge these two techniques and combine their advantages. While utilising the flexibility of data-driven learning, methods like residual learning frameworks and physics-informed neural networks (PINNs) guarantee that machine learning models retain their physical consistency. For sophisticated devices where complete physical modelling is impractical or data is scarce, these hybrid models provide a well-balanced solution by preserving simulation accuracy, enhancing generalisation, and facilitating quicker compact model development. Hybrid techniques are becoming more and more recognised as the most practical means of achieving accurate, comprehensible, and effective device modelling as semiconductor technology continues to advance.

Hybrid methodologies enhance the accuracy and physical reliability of models by integrating domain-specific physics with data-driven learning. Several strategies are employed to incorporate physical constraints into AI models in these approaches. One of the most prevalent methods is loss function engineering, which involves the introduction of physical laws (e.g., conservation of charge, Poisson's equation) as penalty terms during the training process to direct the model towards solutions that are physically plausible. Alternately, architecture-level embedding may be implemented, such as the development of neural networks that enforce predetermined symmetries, monotonicity, or boundary conditions. In certain instances, post-processing corrections are implemented to modify model outputs in accordance with physical laws; however, this approach is less favoured due to the potential for inconsistencies. These hybrid strategies are essential for the development of reliable, trustworthy AI models for semiconductor applications, particularly in low-data regimes or extrapolative scenarios, as they mitigate the risk of unphysical behaviour.

2) ACTIVE LEARNING FOR OPTIMISED PARAMETER SAMPLING

Active learning strategies have been suggested to reduce the quantity of simulation data necessary for training the parameter-extraction model. In this arrangement, the machine learning algorithm sequentially identifies the most informative device configurations for subsequent simulation, utilising uncertainty or gradient data. This method is particularly advantageous for high-fidelity modelling, when each

simulation point, such as a complete TCAD run, incurs significant cost and time investment

3) TRANSFER LEARNING ACROSS TECHNOLOGICAL NODES

A novel technique is employing transfer learning to modify parameter extraction models for various process nodes or device types. A model developed with BSIM-CMG parameters for 7 nm FinFETs can be refined with a limited dataset from 5 nm GAA FETs. This reusability markedly diminishes the data and temporal demands for compact model creation during the initial stages of the Process Design Kit (PDK).

4) INTERPRETABILITY AND ENFORCEMENT OF CONSTRAINTS

A primary concern in ML-assisted parameter extraction is the physical interpretability of the derived values. Recent research tackles this by including physics-informed constraints into machine learning models or loss functions, so assuring that the output parameters adhere to physically viable ranges and comply with monotonicity or symmetry principles. For example, constraint-aware neural networks can guarantee that extracted threshold voltages rise with increasing channel length, hence maintaining established physical trends.

VII. FUTURE OF SYSTEM INTEGRATION WITH SEMICONDUCTOR DEVICE

As semiconductor device modelling advances, we expect to see more automation and integration of AI-driven workflows, allowing for rapid iteration at the device, circuit, and system level. This technique can establish a seamless integration environment in which system-level input automatically informs device-level changes, effectively closing the design-simulation loop as shown in fig 16. The confluence of ML, physics-informed modelling, and multi-domain simulations portends a future in which semiconductor device modelling and system integration are inextricably linked, allowing designers to create highly optimised, scalable, and flexible systems. This shift to integrated modelling frameworks will be critical for satisfying the demands of next-generation applications ranging from IoT and edge computing to artificial intelligence and quantum computing.

VIII. TRANSLATIONAL CHALLENGES FROM DATA-DRIVEN MODELS TO CIRCUIT LEVEL TOOLS

The incorporation of machine learning (ML) engines into semiconductor modelling workflows presents significant promise, especially in reconciling comprehensive physical simulations at the TCAD level with circuit-level assessments through SPICE. Nonetheless, despite this potential, numerous basic obstacles impede the smooth integration of ML-based models throughout the design process.

As semiconductor devices progress below the 5 nm threshold, conventional compact models—based on threshold voltage, charge dynamics, or surface potential equations—fail to effectively represent the intricate physics of nanoscale

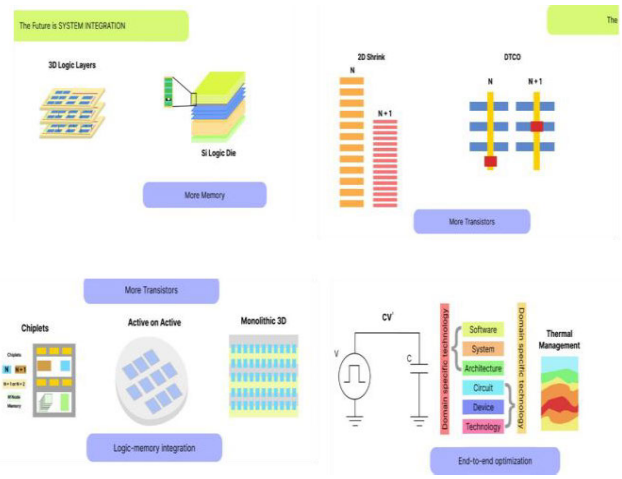

FIGURE 16. Unleashing the future of innovation of semiconductor technologies.

functionality, material interfaces, and electrothermal phenomena. The simultaneous introduction of heterogeneously integrated components and operation in harsh settings hampers model building and parameter extraction. In this context, machine learning approaches have emerged as valuable alternatives, capable of directly learning device behaviour from extensive volumes of TCAD or experimental data. By using the nonlinear approximation capabilities of neural networks, machine learning models circumvent the necessity for explicit physical equations, providing rapidity, adaptability, and data-driven calibration across extensive parametric ranges.

Translating such machine learning models into practical, SPICE-compatible compact forms is complex. In contrast to traditional analytical models, machine learning representations lack standardisation and are frequently trained with architecture-specific data pipelines. This presents challenges when integrating them into established design toolchains, especially where Verilog-A implementations necessitate continuity, differentiability, and reliable convergence behaviour. Neural networks, in their unrefined state, may produce non-smooth outputs or numerical instability when implemented in SPICE solvers, particularly in sensitive analogue or RF circuits.

A significant concern is the deficiency in interpretability and physical uniformity. Although machine learning models can interpolate effectively inside their training domains, they may produce unphysical outcomes when operating outside of these domains, particularly in edge instances or under extrapolated conditions. Conventional models provide designers with straightforward parameter settings and physics-based boundary-aware formulations, which are frequently lacking in black-box machine learning models. It is crucial for ML-based compact models to maintain established trends, such as the monotonic relationship between threshold voltage and channel length, to foster design confidence and model validation.

Moreover, the intricacy of converting machine learning architectures into compact models exacerbates the modelling challenge. In contrast to physics-based methodologies that have gained from years of tool standardisation and PDK integration, machine learning models are devoid of cohesive workflows for data pretreatment, training, validation, and code creation. The difficulty is exacerbated by the lack of benchmark datasets, open-source machine learning exploratory data analysis pipelines, and industry-standard verification methods, hindering the broad comparison and implementation of machine learning models.

A further limitation is the paucity of data regarding failure modes, long-term reliability, and process variability. Machine learning models frequently depend on supervised learning from large datasets; yet, evidence illustrating deterioration mechanisms such as bias temperature instability, hot carrier effects, or self-heating is scarce. This constrains their utility in modelling ageing or forecasting lifespan under diverse environmental situations. Although methodologies like transfer learning and physics-informed learning help alleviate data scarcity, they remain in the exploratory phase and have not achieved widespread industry adoption.

Not withstanding these obstacles, initiatives are in progress to develop hybrid modelling methodologies that integrate data-driven elements with physics-based limitations. Models that include physical laws as regularisation terms or incorporate analytical sub-blocks inside learnt frameworks demonstrate potential in enhancing generalisation and ensuring physical consistency. Furthermore, initial classification frameworks, including i-type (entirely data-driven), ii-type (physics-regularized), and iii-type (physics-informed hybrid) models, have been suggested to facilitate model selection according to complexity, accuracy, and deployment requirements.

Despite their advantages, AI-driven models have a number of significant drawbacks that prevent them from being used more widely in semiconductor device modelling. Their inability to generalise is a significant problem; these models frequently only function well in the limited domain of their training data and find it difficult to anticipate behaviour under novel operating conditions, material systems, or device architectures. Their significant data dependency is made worse by the fact that training successful models necessitates vast amounts of diverse, high-quality datasets, which are sometimes unfeasible or prohibitively expensive to collect through experimental measurements or TCAD simulations. Furthermore, AI models—deep neural networks in particular—function as ''black boxes,'' providing little to no explanation of the physical processes that underlie their predictions. This makes it difficult for subject-matter experts to understand and accept these models. Additionally, they frequently disregard basic physical principles, which could lead to nonphysical predictions like negative voltages or currents. Additionally, training these models requires a significant amount of computing power, which not all research environments can provide. Lastly, these models' lack of transparency makes it difficult for regulators to validate them in safety-critical applications, and their limited transferability across various technological nodes or applications necessitates extensive retraining. These drawbacks highlight the increasing interest in hybrid strategies that combine data-driven methods with physical knowledge to improve generalisation and dependability. Training AI models with minimal experimental data is difficult, especially for innovative device architectures like 2D-material FETs, tunnel FETs, and quantum-dot-based devices. Due to initial research, fabrication difficulty, and high measuring infrastructure costs, emerging technologies generally lack comprehensive datasets. Due to scarce data, models trained on it tend to overfit and learn noise or specialised patterns rather than generalizable trends. The lack of labelled data prevents the model from capturing unusual or nonlinear behaviours like short-channel effects or quantum confinement, which are essential for proper device modelling. Novel device experimental data may be noisy or inconsistent due to process variability or immature fabrication procedures, affecting model training and validation. Deep learning architectures, which need huge datasets to operate well, are likewise constrained by data availability. Traditional image or text data augmentation methods cannot be applied to semiconductor data without risking physical inconsistencies. To construct viable AI frameworks for next-generation semiconductor devices, physics-informed learning, transfer learning from established device models, and synthetic data generation (e.g., via TCAD or generative models) are needed.

The future of ML-integrated modelling resides in the creation of cohesive, scalable, and automated frameworks capable of facilitating end-to-end modelling from TCAD-level learning to SPICE-level simulation. Standardised datasets, open-model benchmarks, and collaborative validation initiatives between academics and industry will be crucial in cultivating trust and interoperability. Table 2 summarizes a brief comparison between the conventional and AI –Based Modelling Approaches. As these issues are resolved, machine learning is anticipated to significantly influence the development of compact modelling procedures for next-generation semiconductor technologies.

IX. MODEL VALIDATION IN AI-BASED DEVICE MODELING

Making sure that AI-generated models are physically reliable and accurate in their predictions is a very important part of using them to simulate semiconductor devices. Model validation that works well connects predictions based on data with the physics of the device itself. Several research have shown that validation is often done by comparing the outputs of an AI model to experimental data or well-known TCAD simulation results. People often use quantitative indicators like Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2) to judge performance. K-fold cross-validation also makes sure that the model works well on many types of devices and in different settings. It is just as crucial to check for physical

TABLE 2. Comparison of conventional vs. AI-based modeling approaches.

Aspect	Conventional (Physics-Based) Models	AI-Based (Data-Driven) Models
Core Principle	Based on fundamental physical laws (e.g., Poisson's, Schrödinger's equations)	Learns behavior from data using statistical or machine learning techniques
Model Transparency	High (white-box); interpretable and explainable	Low (black-box); limited interpretability
Accuracy	High within well-understood regimes	High when trained on sufficient, high-quality data
Generalizability	Good across devices with known physics	Limited; weak performance on unseen configurations
Data Requirements	Low; needs only physical parameters	High; requires large, diverse datasets
Computational Cost	High for complex geometries or quantum effects	High during training; low during inference
Adaptability to Novel Devices	Requires reformulation of physical models	Flexible; can adapt if trained on new data
Noise Robustness	Robust due to physical consistency	Sensitive to noise and measurement errors

consistency, including making sure that the trends in threshold voltage, subthreshold swing, and saturation behaviour are correct under varied operating settings. Adding these kinds of validation techniques not only makes the model more believable, but it also makes the semiconductor research community more likely to use AI methods.

X. OPEN CHALLENGES AND FUTURE RESEARCH DIRECTIONS IN AI-DRIVEN DEVICE MODELING

Numerous open challenges and prospective research directions remain in order to fully realise the potential of AI-driven semiconductor device modelling, despite recent progress. The development of physics-informed machine learning (PIML) frameworks that incorporate domain knowledge with data-driven models is a critical need. This approach ensures physical consistency and enhances prediction accuracy. Furthermore, the discipline must confront the challenge of data scarcity, particularly for emerging devices, by employing synthetic data generation, transfer learning, and active learning strategies to mitigate its dependence on extensive experimental datasets. Another critical objective is to enhance the interpretability of models in order to cultivate trust among regulatory bodies and device engineers. Methods such as symbolic regression and explainable AI (XAI) may offer valuable insights into learnt relationships. Additionally, there is an increasing need for effective models that are scalable and capable of effectively connecting device- and system-level simulations. Additionally, in order to guarantee reproducibility and equitable comparisons among various methodologies, it is imperative to benchmark and standardise datasets, evaluation metrics, and modelling protocols.

AI-driven semiconductor device modelling faces various obstacles despite progress. Future research should focus on physics-informed machine learning (PIML) models that smoothly integrate fundamental physical laws into data-driven frameworks to match device behaviour. Process-induced variability and measurement noise in semiconductor production are another major issue. Under these uncertainties, current AI models struggle to generalise, making predictions untrustworthy. Noise-aware learning, uncertainty quantification (e.g., Bayesian neural networks), and ensemble models with reliability estimates should be explored in future research. Data scarcity must be overcome, especially for novel device architectures, via transfer learning, active learning, or high-fidelity simulations.

Finally, it will be imperative to establish a more effective partnership between the semiconductor and AI communities in order to ensure that algorithmic advancements are in accordance with the changing requirements of circuit design, process modelling, and device physics.

XI. CONCLUSION

A systematic review of new device models covers both machine-learning-based and conventional device-physics engines. Near-threshold modelling, growing complexity, atomic-scale physics, reliability characterisation, heterointegration in ICs, and extreme operating environments are challenges that conventional physics-driven models—such as threshold voltage-based, surface potential-based, and charge-based models—face. As sophisticated devices continue to scale, these difficulties impede the accurate creation of models and the effective extraction of parameters.

Regardless of the intricacy of their physics, machine learning (ML) techniques have emerged as potent instruments for modelling semiconductor devices in order to overcome these constraints. These models can learn from large amounts of simulations and experimental data by utilising the universal approximation capabilities of artificial neural networks. Through a methodical process that includes parameter determination (step 1), dataset preparation (step 2), algorithm selection (step 3), Verilog-A coding (step 4), and circuit simulation (step 5), this approach makes it possible to construct generic MOSFET compact models.

The inference speed of AI-based models is a significant advantage. Once trained, they can generate results at a rate

that is orders of magnitude faster than traditional TCAD solvers, making them well-suited for real-time or large-scale applications. Significant computational resources, such as GPU or HPC clusters, are frequently required for hours to days of training in deep learning models, particularly those that employ large datasets or physics-informed architectures like PINNs. Conversely, conventional physics-based solvers, such as TCAD tools, necessitate no training, despite their computationally intensive nature per simulation run. For instance, in TCAD, the simulation of a device with quantum effects may require several minutes to hours per configuration, whereas a trained neural network can predict outcomes in milliseconds. Consequently, AI models facilitate quicker implementation by transferring the computational expense from runtime to pre-training when they are integrated into design workflows.

This new approach can greatly improve TCAD-to-SPICE workflows, which are becoming more and more important for design-technology co-optimization (DTCO), and it provides a potential option for semiconductor modelling. In contrast to industry-standard models, there are still a number of obstacles to overcome, including the absence of industry-wide standards, the difficulty of implementing algorithms, the creation of sub-models, and maintaining physical consistency.

In order to get over these challenges, developments have been divided into three types of models: i-type, ii-type, and iii-type. These models are differentiated by the datasets and current expressions they use. These developments increase performance by utilising improved algorithms or existing information. Accelerating the design, modelling, and optimisation of large-scale circuit systems is anticipated to be greatly aided by the development of precise, unified, and automated machine-learning-based compact models (MLCMs). Moreover, hybrid methods that combine data and physics could bridge the gap between theoretical developments and production needs.

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